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Exceptional Evolutionary Expansion of Prefrontal Cortex in Great Apes and Humans

Highlights

- Great ape and human prefrontal expansion are evolutionarily derived features
- Great apes and humans are specialized to favor executive cognitive function
- This exceptional prefrontal expansion is likely related to heterochronic remodeling

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In Brief

Smaers et al. report that great ape and human prefrontal cortex expansion are evolutionarily specialized features of cortical organization, favoring executive cognitive function within distributed networks. This pattern of cortical reorganization is likely related to heterochronic changes that prolong prefrontal development.



Exceptional Evolutionary Expansion of Prefrontal Cortex in Great Apes and Humans

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SUMMARY

One of the enduring questions that has driven neuroscientific enquiry in the last century has been the nature of differences in the prefrontal cortex of humans versus other animals [1]. The prefrontal cortex has drawn particular interest due to its role in a range of evolutionarily specialized cognitive capacities such as language [2], imagination [3], and complex decision making [4]. Both cytoarchitectonic [5] and comparative neuroimaging [6] studies have converged on the conclusion that the proportion of prefrontal cortex in the human brain is greatly increased relative to that of other primates. However, considering the tremendous overall expansion of the neocortex in human evolution, it has proven difficult to ascertain whether this extent of prefrontal enlargement follows general allometric growth patterns, or whether it is exceptional [1]. Species' adherence to a common allometric relationship suggests conservation through phenotypic integration, while species' deviations point toward the occurrence of shifts in genetic and/or developmental mechanisms. Here we investigate prefrontal cortex scaling across anthropoid primates and find that great ape and human prefrontal cortex expansion are non-allometrically derived features of cortical organization. This result aligns with evidence for a developmental heterochronic shift in human prefrontal growth [7, 8], suggesting an association between neurodevelopmental changes and cortical organization on a macroevolutionary scale. The evolutionary origin of non-allometric prefrontal enlargement is estimated to lie at the root of great apes (~19–15 mya), indicating that selection for changes in executive cognitive functions characterized both great ape and human cortical organization.

RESULTS

Phylogenetic analysis of covariance [9] reveals that a multi-grade isometric model (dividing humans, great apes, and other pri-

mates) provides a significantly better statistical fit to prefrontal scaling than a one-grade allometric model (Figures 1, 2 and 3, Tables S1 and S2). This applies to the comparison of prefrontal cortex with several other cortical areas that are functionally and neurobiologically linked to it. Comparing cortical areas that are neurobiologically linked ensures that the measure of relative prefrontal size accounts for the hierarchical nature of neural information processing (see [10] and Supplemental Information for more details). We consider only datasets of prefrontal cortex size in primates that have been collected based on cytoarchitectonic criteria and comprise information for more than five species [5, 11] (Figure 1, Table S1; see Supplemental Information for more details). These criteria lead to the selection of two datasets, designated as Brodmann and Smaers datasets hereafter. Phylogenetic ANCOVA was performed by generalizing the standard generalized least-squares procedure to including phylogenetic variance-covariance in combination with additional indicator variables that describe group membership (even when a group consists of a single species; see [9] and Supplemental Information for a detailed description of this approach). Because this implementation of phylogenetic ANCOVA uses standard least-squares procedures only, it provides unbiased results irrespective of sample size (see Supplemental Information for more details). Because statistical power is a simple function of the observed p value, the significant results denote that the tests presented here have high observed power (see Supplemental Information for more details).

To elucidate when in evolutionary time episodes of enlargement in prefrontal cortex occurred, we further used three different evolutionary modeling approaches. These methods explore differences among groups directly from the data (i.e., without a priori group allocation). Results demonstrate that prefrontal cortex exhibits separate instances of exceptional expansion in the hominoid (~30–19 mya), hominid (i.e., great ape and human) (~19–15 mya), human-chimpanzee (~8–6 mya), and human (~6–0 mya) ancestral lineages when compared to different brain structure scaling variables. Ancestral state and rate estimation results (Figures 2 and 3) visualize best estimates of how prefrontal cortex has changed along individual lineages of the primate tree, and best-fit regime configurations highlight sets of lineages (“regimes”) that indicate similar trait values for relative prefrontal cortex size in addition to where shifts between regimes occurred in phylogenetic space. These results demonstrate that exceptional prefrontal expansion relative to frontal

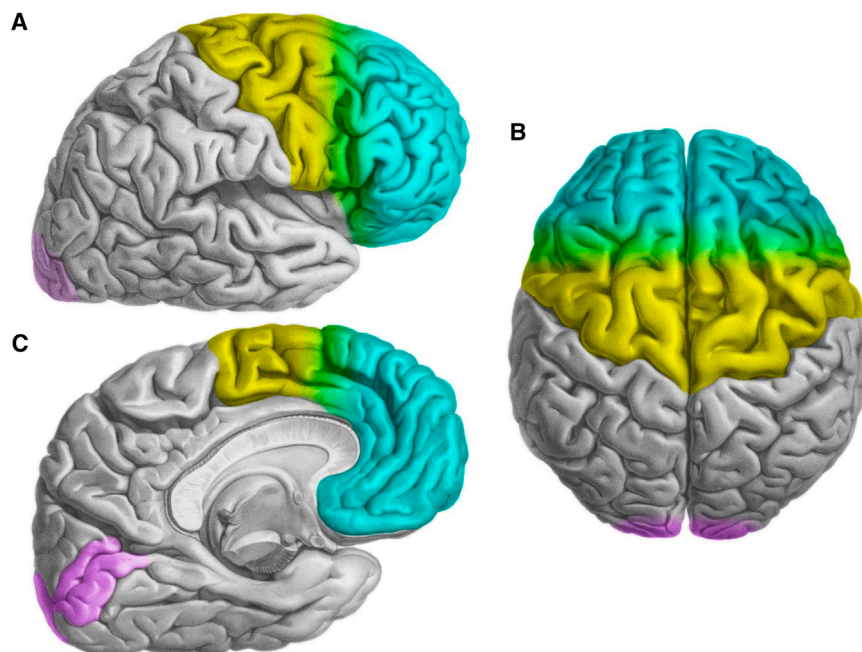


Figure 1. Overview of Brain Regions

Lateral (A), dorsal (B), and medial (C) views of the human brain illustrating the regions under consideration. Red illustrates the primary visual cortex, yellow the frontal motor areas, and blue the prefrontal cortex. The green area depicts a margin of uncertainty in the location of the cytoarchitectonic border between frontal motor areas and prefrontal cortex when using the prefrontal delineation approach proposed by Smaers et al. [11]. This approach considers a series of cumulative volumes along the frontal pole as a proxy for prefrontal cortex volume. This approach allows for the collection of a valid proxy for prefrontal cortex volume in a wide sample of species but results in an underestimation of putative prefrontal expansion in great apes and humans (see [Supplemental Information](#) for more details). The Brodmann data provide a more accurate measure of prefrontal cortex size but comprise a more limited comparative sample. Figure adjusted from Foville [40]. See also [Tables S1 and S3](#).

motor areas is estimated to have occurred in the ancestral hominoid lineage, expansion of prefrontal cortex relative to primary visual cortex is estimated to have occurred in the ancestral lineages of great apes and humans, and prefrontal expansion relative to other heteromodal association areas is estimated to have occurred in the human lineage. The statistical effect size of these evolutionary models is high ($\sqrt{n}\phi \gg 1$, $\beta/\sqrt{\gamma} \gg 2$), rendering strong support for the occurrence of these evolutionary shifts in relative prefrontal enlargement (see more details in [Supplemental Information](#)). Bootstrap analysis further supports this conclusion by demonstrating high support for the estimated trait shifts ([Figures 2, 3, and S1](#), see [Supplemental Information](#) for more details). High effect size is also supported by the fact that the same result is obtained using four different methods (phylogenetic ANCOVA, ancestral estimation, rate estimation, and multi-regime OU model fitting) and five different model assumptions (see [Supplemental Information](#) for more details).

DISCUSSION

Whether or not human prefrontal cortex expansion is predictable from common rules for primate brain allometric scaling bears on the fundamental question of the extent to which human cortical organization can be accounted for solely by genetic and developmental patterns shared with other primates. Although many biological systems are primarily integrated (i.e., different elements of the system change in a coordinated manner), deviations from integration are common and are tied to genetic, developmental, and/or functional shifts in a species' bauplan (e.g., heterochronies [12]). These modifications shape the direction of trait variation on a macroevolutionary scale [12] and are thus fundamental drivers of biological diversity. To understand whether an event aligns with or deviates from integration, a standard approach has been to investigate allometric

conservation of traits [13]. Species' adherence to the allometric relationship hereby suggests conservation through phenotypic integration, while species' deviations from the allometric relationship points toward a genetic or developmental shift.

Our results support the conclusion that great ape and human prefrontal expansion are evolutionary specializations of cortical organization that cannot be explained solely by allometric trends. These results align with studies demonstrating that human prefrontal cortex is relatively specialized compared to other primates. Human prefrontal neurons are characterized by higher dendritic branch complexity and synaptic spine density compared to other heteromodal association regions and nonhuman primate prefrontal cortex [14], and regions of the human prefrontal cortex contain more neuropil space than in other great apes [15]. Such neuroanatomical differences may be linked to human-specific increased transcriptional complexity [16] and alterations in the regulation of gene expression [17]. These gene regulatory changes have been suggested to have arisen through heterochronic remodeling of the developmental patterns that underpin human prefrontal growth [7, 8]. In general, across different biological systems, heterochrony (i.e., changes in the rate and timing of developmental patterns) has been shown to underlie deviations from phenotypic integration by altering genetic, developmental, and/or functional effects and leading to changes in the direction of trait variation on a macroevolutionary scale [12]. Indeed, heterochrony has previously been suggested to be an important driver of volumetric reorganization in the mammalian brain [18]. In the case of human prefrontal cortex, developmental changes in mRNA expression have been characterized as comparatively prolonged, or neotenic [8], and shown to have evolved at an accelerated rate relative to chimpanzees, macaques, and human non-cortical regions [7]. In particular, peak expression for synapse-associated genes is delayed to approximately 5 years after birth in humans, compared to a few months after birth in chimpanzees and macaques [19].

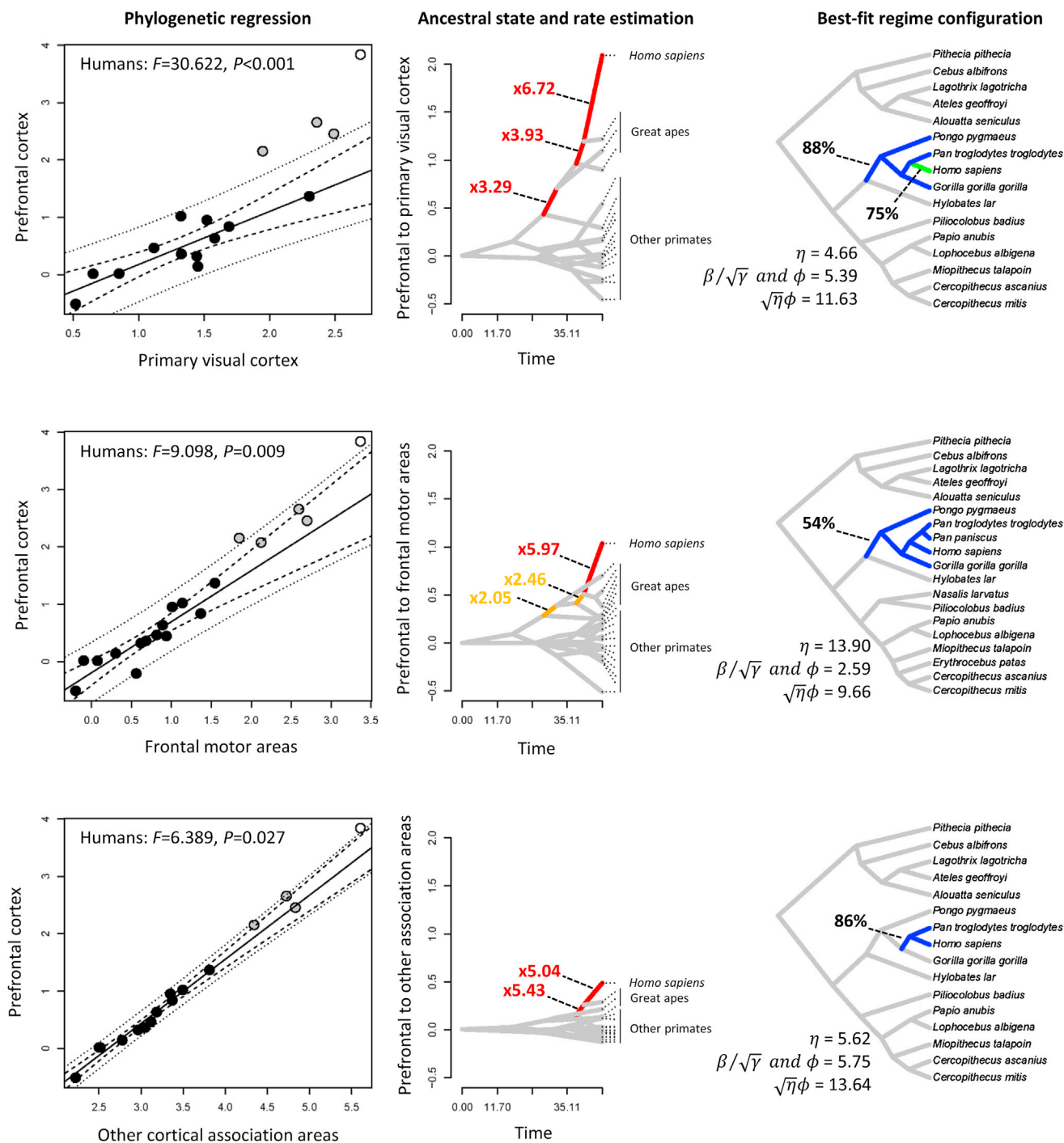


Figure 2. Evolutionary Modeling of Prefrontal Expansion using the Smaers Data

Phylogenetic regressions of log prefrontal cortex size against log size of other cortical areas. Prefrontal data from Smaers et al. [11]. Slopes, confidence intervals (dashed line), and prediction intervals (dotted line) [9] are depicted based on the non-great ape sample. Data points with a white background represent human values, those with a gray background great ape values. F and p values indicate the significance of a phylogenetic ANCOVA testing for intercept differences between humans and other primates (see also Smaers and Rohlf [9], Supplemental Information on the use and interpretation of phylogenetic ANCOVA, and Table S2 for more detailed results). Ancestral state and rate estimation plots visualize lineage-specific phenotypic change across time in an ancestral phenogram. Branches are colored according to the extent to which their rate of evolution is larger than expected based on a neutral constant-rate model of evolution (orange = 2–3 times larger; red = more than 3 times larger). Best-fit regime configurations highlight branches with a similar trait value as estimated by a least-squares lasso procedure using a phylogenetic Bayesian information criterion (BIC) [37]. Colors differentiate between significantly different regimes (“regime” is here defined as a

(legend continued on next page)

Furthermore, cortical myelination is completed at the age of sexual maturity for chimpanzees and macaques, while human axonal maturation extends well into the third decade of life [20]. Genetic and developmental studies thus provide evidence for a heterochronic shift in human prefrontal growth. Our results are consistent with these findings, suggesting that such neurodevelopmental changes are associated with cortical reorganization on a macroevolutionary scale.

In addition to exceptional prefrontal expansion in humans, we also demonstrate that great apes show prefrontal enlargement that deviates significantly from the expected allometric scaling pattern of cortical integration in other primates. This result is congruent with the general consensus from comparative psychology and primatology that the level of cognitive abilities in great apes (e.g., higher levels of self-control [21] and cultural traditions [22]) is distinct from that in other primates. Such evolutionary changes in brain organization of great apes relative to other primates provides a possible springboard for future broad comparative genetic and developmental investigations into the mechanisms that shape these neurobiological changes over time.

These results also align with previous suggestions that great ape and human neocortical expansion is primarily due to enlargement of cortical association areas, whereas koniocortex (e.g., primary sensory cortices) scales more closely with body size [6, 23]. Previous assertions that “the size of human frontal lobes, and of specific frontal regions, is as expected relative to the size of other brain structures” [24] do not have support according to our analyses. To further underline this issue, Figure 4 plots size measurements of prefrontal cortex relative to other brain structures for the Brodmann dataset. The isometric relationship of prefrontal cortex to primary visual cortex and frontal motor areas is evident in the proportional size changes in the non-great ape sample. In the human brain, however, prefrontal cortex expands exponentially, while primary visual cortex and frontal motor areas remain in line.

Previous conclusions that human prefrontal cortex size or neuron numbers are only as large as predicted for a scaled-up monkey brain [24, 25] can be explained by three fundamental factors. First, previous studies have not employed statistical procedures to test for significant differences in the intercept among subgroups before interpreting allometry [9]. Second, previous allometric studies have used datasets that are not adequate for the interpretation of comparative differences in prefrontal volumes or neuron numbers. Although cortical areas are defined by functional, connectional, and cytoarchitectonic criteria [5], not by gross anatomical or external morphological characteristics [26], previous allometric studies have used datasets that define prefrontal cortex as all cortex anterior to corpus callosum [24, 25], likely because it is a proxy that is simple and easily applied. However, human prefrontal cortex extends further along the caudal axis of the frontal pole than in chimpanzees and other primates [6], making this delineation result in a disproportionate underestimation of prefrontal cortex in humans relative to that in non-human primates,

thus rendering the measure inaccurate for the purposes of volumetric comparison across species. Third, the investigation of putative expansion of a cortical area is commonly evaluated relative to the size of the rest of the cerebral cortex or the rest of the brain [24]. This approach does not account for the functional and anatomical underpinnings of neural information processing [10]. Information ascends initially through primary sensory areas, after which it is integrated in supplementary sensory and temporo-parietal association areas to form mental representations. Prefrontal cortex subsequently exerts control over the manipulation of and changes in these mental representations [4]. Here we show that an evaluation of prefrontal cortex size that accounts for this hierarchical nature of information processing unequivocally indicates the exceptional expansion of prefrontal cortex in great apes and humans (Figures 2 and 3, Table S2). Nonetheless, when using the phylogenetic statistics and evolutionary modeling methods employed in the current study, even the more coarse comparison of prefrontal size to the size of the rest of the cortex (used in previous work to argue that human prefrontal cortex size is as expected for a scaled-up monkey brain [24, 25]) yields a marginally significant result for gray matter ($F = 4.104$, $p = 0.061$) and a significant result for white matter ($F = 5.981$, $p = 0.027$) for the Smaers dataset (which by design provides an underestimate of human prefrontal expansion, see details in Supplemental Information) and a strongly significant result for the Brodmann dataset ($F = 18.921$, $p = 0.007$).

The functional implications of exceptional human prefrontal expansion has previously been interpreted as a potential neural basis for human behavioral and cognitive distinctiveness. One possibility is that extraordinary prefrontal enlargement in great apes and humans is due to the evolution of novel cortical areas. Although an impressive body of work suggests that the basic map of prefrontal areas is largely homologous in Old World monkeys and humans [27], some evidence suggests that the human prefrontal cortex may contain new regions. Brodmann [5] found no nonhuman homologs for areas 45, 46, and 47 (but see work by Petrides and Pandya [28]), and recent research indicates the possibility of major changes in neurogenesis and neural migration that may underpin changes in the distribution of cell types in human prefrontal cortex [29]. More research is needed to provide a definitive answer in this regard. It is, however, a distinct possibility that, rather than being characteristic of human prefrontal cortex evolution, the addition of novel cortical areas may be more characteristic of early primate evolution. The dorsolateral prefrontal cortex, for example, together with a suite of other cortical (superior temporal sulcus, inferior temporal, posterior parietal, ventral and dorsal premotor) and thalamic (dorsal pulvinar) areas have been shown to be functionally and cytoarchitectonically distinct in primates compared to other mammals [30].

Another, though not mutually exclusive, possibility is that new specializations of the great ape and human prefrontal cortex comprise a shift in their network organization with other regions by means of connectional invasion. This evolutionary-developmental process occurs when hypertrophied areas invade targets they do not typically innervate in other species and/or increase

cluster of branches with a similar trait value). Bootstrap support is indicated at the ancestral branch of each regime. Effect size is indicated using different measures. Ho and Ané [38] suggest $\beta/\sqrt{\gamma} > 2$ as a valid indicator of high effect size, whereas Cressler et al. [39] propose $\sqrt{\eta\phi} \gg 1$. According to every proposed measure, analyses presented here demonstrate high effect size, and thus high observed power. See Tables S2–S6 and Figures S1 and S2 for more details and supplemental results. Figure S1 and Table S5 demonstrate that similar results are obtained for the analysis of white matter.

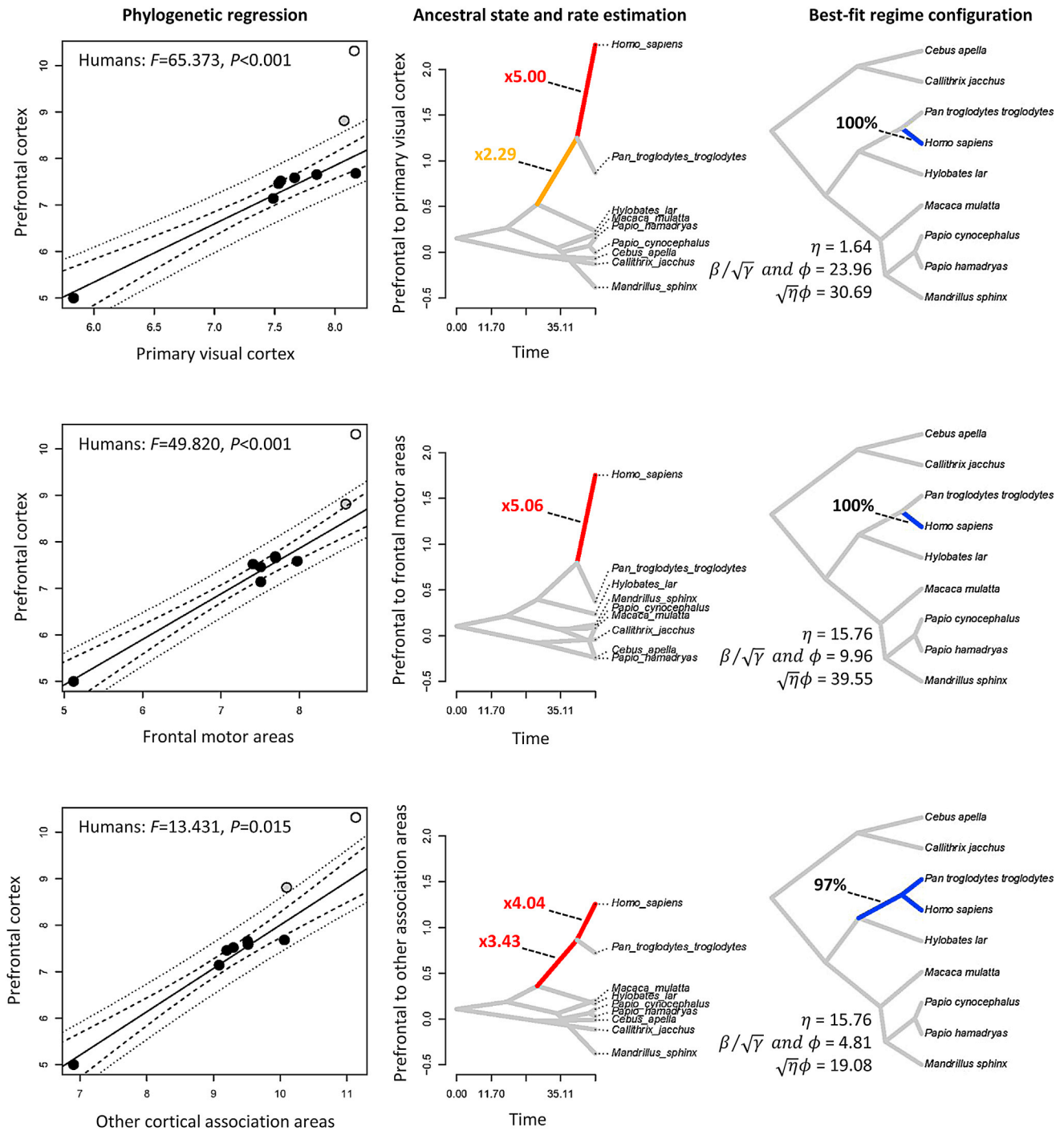


Figure 3. Evolutionary Modeling of Prefrontal Expansion using the Brodmann Data

As in Figure 2, data from Brodmann [5]. See also Figure S1 and Tables S2–S6.

target innervation relative to the ancestral condition [31]. Such new connections may displace others, causing the hypertrophied areas to exert more influence over information processing. This possibility refocuses the characterization of human brain uniqueness toward a distributed neural network in which the prefrontal cortex plays a dominant role. A likely candidate for such a distributed network is the prefronto-cerebellar system. Prefron-

tal projecting cerebellar lobules have been shown to demonstrate a hominoid/hominid grade shift in size [32] similar to that observed in the prefrontal cortex, to have co-evolved with the prefrontal cortex in great ape and human lineages [33], and to underlie a range of behaviors often associated with human behavioral distinctiveness (e.g., language and executive function [34]). Other distributed networks that may be of particular

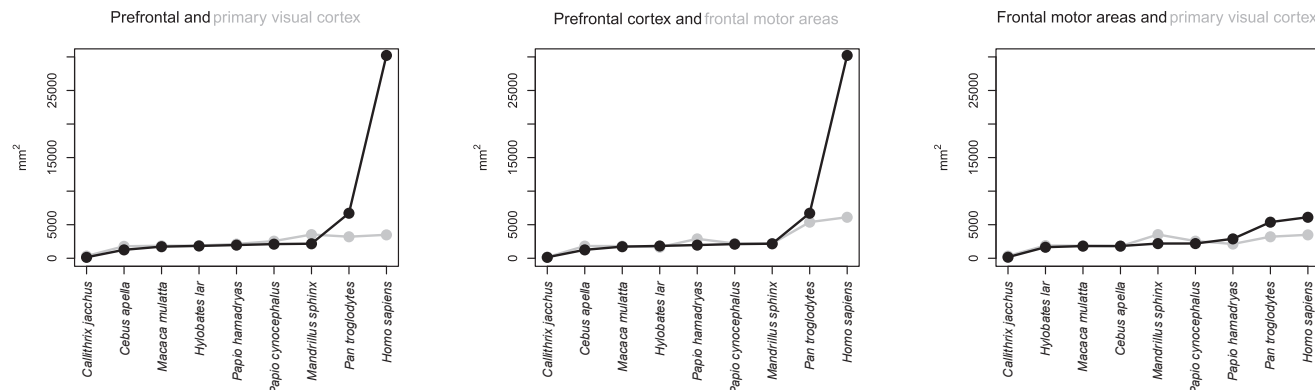


Figure 4. Expansion of Prefrontal Size in the Human Lineage

Plot of prefrontal size and the size of other brain structures. Species are rank ordered according to the size of the first variable in the comparison (highlighted in black). Data from Brodmann [5]. See also Figure S2.

interest in this context are the prefronto-parietal [35] and prefronto-temporal [36] pathways. Considering the exceptional enlargement of prefrontal cortex in great apes and humans, this would suggest that cortical organization in humans and great apes is evolutionary specialized to favor prefrontal cortex function within distributed networks.

We conclude that both human and great ape brain evolution is characterized by non-allometrically derived changes in cortical organization comprising the exceptional expansion of prefrontal cortex. This expansion should be contextualized as part of the elaboration of a large-scale network that involves prefrontal cortex, temporo-parietal cortex [6, 10], and cerebellar hemispheres [32]. Considering that this network is thought to have arisen early in primate evolution [30], great ape and human brains can be considered as extreme (non-allometrically derived) versions of a primate template of cortical organization. The expansion of human prefrontal cortex significantly exceeds the enlargement in other heteromodal association areas, suggesting that human evolution has been characterized by selection for changes in executive functions mediated by this cortical region. The congruence between evidence for heterochronic remodeling of human prefrontal growth with the macroevolutionary expansion of human prefrontal cortex further suggests that the same developmental mechanisms that have been shown to be fundamental drivers of diversity across different biological systems in mammals (e.g., heterochrony) [12] shape primate neurobiological diversity in a similar way.

EXPERIMENTAL PROCEDURES

Data

We consider only the available datasets of prefrontal cortex size in primates that have been collected based on cytoarchitectonic criteria and comprise information for more than five species [5, 11]. These datasets, collected by Brodmann and by Smaers and colleagues, differ in the breadth of the comparative sample (13 versus 19 species, respectively), the nature of the measurement (mm² versus mm³), and the cytoarchitectonic criteria that were employed (granular and agranular cortex versus volumetric bootstrapping along the frontal pole [11] relative to the cytoarchitectonic border between areas 3 and 4). These differences are such that the Brodmann data provide a more accurate delineation of prefrontal cortex size but a more modest comparative sample, whereas the Smaers data provide a larger comparative sample but a proxy for prefrontal

cortex volume that underestimates any putative prefrontal expansion in great apes and humans (see Supplemental Information for more details).

Analysis

Phylogenetic ANCOVA was used to test for differences in intercepts and slopes among subgroups (humans versus great apes versus other primates). Such formal tests are required to evaluate whether slopes and intercepts are homogeneous in all subgroups of the sample before interpreting allometry. We used an implementation of phylogenetic ANCOVA that uses standard least-squares procedures only [9], thus ensuring unbiased calculation of regression parameters irrespective of sample size. This method further uses the standard approach of degrees of freedom to penalize for model parameterization to guard against overfitting (see Supplemental Information for more details and additional tests that exemplify this feature).

Best-fit evolutionary scenarios were obtained with least-squares multi-regime Ornstein-Uhlenbeck modeling procedures in combination with a conservative model selection criterion (phylogenetic BIC) to avoid overfitting [37, 38] (see Supplemental Information for more details and additional tests that exemplify results are robust against overfitting). When data indicate a high effect size, this approach has been shown to provide a high power even for sample sizes as few as ten taxa [39]. All analyses presented here indeed show high effect size (see Figures 2 and 3 and Supplemental Information for more details). Ancestral estimates and branch-specific rates were obtained using two different multi-rate models of evolution, both of which yielded equivalent results (see Supplemental Information for more details).

SUPPLEMENTAL INFORMATION

Supplemental Information includes two figures, six tables, and Supplemental Experimental Procedures and can be found with this article online at <http://dx.doi.org/10.1016/j.cub.2017.01.020>.

AUTHOR CONTRIBUTIONS

Research Design, J.B.S. and C.C.S.; Statistical Analyses, J.B.S. and A.G.-R.; Writing – Original Draft, J.B.S. and A.N.P.; Writing – Review and Editing, all authors.

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REFERENCES

- Sherwood, C.C., and Smaers, J.B. (2013). What's the fuss over human frontal lobe evolution? *Trends Cogn. Sci.* 17, 432–433.
- Gabrieli, J.D.E., Poldrack, R.A., and Desmond, J.E. (1998). The role of left prefrontal cortex in language and memory. *Proc. Natl. Acad. Sci. USA* 95, 906–913.
- Schacter, D.L., Addis, D.R., and Buckner, R.L. (2007). Remembering the past to imagine the future: the prospective brain. *Nat. Rev. Neurosci.* 8, 657–661.
- Miller, E.K., and Cohen, J.D. (2001). An integrative theory of prefrontal cortex function. *Annu. Rev. Neurosci.* 24, 167–202.
- Brodmann, K. (1909). Vergleichende lokalisationslehre der grosshirnrinde in ihren prinzipien dargestellt auf grund des zellenbaues (Leipzig: Barth).
- Glasser, M.F., Goyal, M.S., Preuss, T.M., Raichle, M.E., and Van Essen, D.C. (2014). Trends and properties of human cerebral cortex: correlations with cortical myelin content. *Neuroimage* 93, 165–175.
- Somel, M., Liu, X., Tang, L., Yan, Z., Hu, H., Guo, S., Jiang, X., Zhang, X., Xu, G., Xie, G., et al. (2011). MicroRNA-driven developmental remodeling in the brain distinguishes humans from other primates. *PLoS Biol.* 9, e1001214.
- Somel, M., Franz, H., Yan, Z., Lorenc, A., Guo, S., Giger, T., Kelso, J., Nickel, B., Dannemann, M., Bahn, S., et al. (2009). Transcriptional neoteny in the human brain. *Proc. Natl. Acad. Sci. USA* 106, 5743–5748.
- Smaers, J.B., and Rohlf, F.J. (2016). Testing species' deviation from allometric predictions using the phylogenetic regression. *Evolution* 70, 1145–1149.
- Passingham, R.E., and Smaers, J.B. (2014). Is the prefrontal cortex especially enlarged in the human brain allometric relations and remapping factors. *Brain Behav. Evol.* 84, 156–166.
- Smaers, J.B., Steele, J., Case, C.R., Cowper, A., Amunts, K., and Zilles, K. (2011). Primate prefrontal cortex evolution: human brains are the extreme of a lateralized ape trend. *Brain Behav. Evol.* 77, 67–78.
- Goswami, A., Smaers, J.B., Soligo, C., and Polly, P.D. (2014). The macro-evolutionary consequences of phenotypic integration: from development to deep time. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 369, 20130254.
- Finlay, B.L., and Darlington, R.B. (1995). Linked regularities in the development and evolution of mammalian brains. *Science* 268, 1578–1584.
- Bianchi, S., Stimpson, C.D., Bauernfeind, A.L., Schapiro, S.J., Baze, W.B., McArthur, M.J., Bronson, E., Hopkins, W.D., Semendeferi, K., and Jacobs, B. (2013). Dendritic morphology of pyramidal neurons in the chimpanzee neocortex: regional specializations and comparison to humans. *Cereb. Cortex* 23, 2429–2436.
- Spocter, M.A., Hopkins, W.D., Barks, S.K., Bianchi, S., Hehmeyer, A.E., Anderson, S.M., Stimpson, C.D., Fobbs, A.J., Hof, P.R., and Sherwood, C.C. (2012). Neuropil distribution in the cerebral cortex differs between humans and chimpanzees. *J. Comp. Neurol.* 520, 2917–2929.
- Konopka, G., Friedrich, T., Davis-Turak, J., Winden, K., Oldham, M.C., Gao, F., Chen, L., Wang, G.-Z., Luo, R., Preuss, T.M., and Geschwind, D.H. (2012). Human-specific transcriptional networks in the brain. *Neuron* 75, 601–617.
- Muntané, G., Horvath, J.E., Hof, P.R., Ely, J.J., Hopkins, W.D., Raghanti, M.A., Lewandowski, A.H., Wray, G.A., and Sherwood, C.C. (2015). Analysis of synaptic gene expression in the neocortex of primates reveals evolutionary changes in glutamatergic neurotransmission. *Cereb. Cortex* 25, 1596–1607.
- Workman, A.D., Charvet, C.J., Clancy, B., Darlington, R.B., and Finlay, B.L. (2013). Modeling transformations of neurodevelopmental sequences across mammalian species. *J. Neurosci.* 33, 7368–7383.
- Liu, X., Somel, M., Tang, L., Yan, Z., Jiang, X., Guo, S., Yuan, Y., He, L., Oleksiak, A., Zhang, Y., et al. (2012). Extension of cortical synaptic development distinguishes humans from chimpanzees and macaques. *Genome Res.* 22, 611–622.
- Miller, D.J., Duka, T., Stimpson, C.D., Schapiro, S.J., Baze, W.B., McArthur, M.J., Fobbs, A.J., Sousa, A.M., Šestan, N., Wildman, D.E., et al. (2012). Prolonged myelination in human neocortical evolution. *Proc. Natl. Acad. Sci. USA* 109, 16480–16485.
- Osvath, M., and Osvath, H. (2008). Chimpanzee (*Pan troglodytes*) and orangutan (*Pongo abelii*) forethought: self-control and pre-experience in the face of future tool use. *Anim. Cogn.* 11, 661–674.
- van Schaik, C.P., Ancrenaz, M., Borgen, G., Galdikas, B., Knott, C.D., Singleton, I., Suzuki, A., Utami, S.S., and Merrill, M. (2003). Orangutan cultures and the evolution of material culture. *Science* 299, 102–105.
- Passingham, R.E. (1975). Changes in the size and organisation of the brain in man and his ancestors. *Brain Behav. Evol.* 11, 73–90.
- Barton, R.A., and Venditti, C. (2013). Human frontal lobes are not relatively large. *Proc. Natl. Acad. Sci. USA* 110, 9001–9006.
- Gabi, M., Neves, K., Masseron, C., Ribeiro, P.F., Ventura-Antunes, L., Torres, L., Mota, B., Kaas, J.H., and Herculano-Houzel, S. (2016). No relative expansion of the number of prefrontal neurons in primate and human evolution. *Proc. Natl. Acad. Sci. USA* 113, 9617–9622.
- Sherwood, C.C., Broadfield, D.C., Holloway, R.L., Gannon, P.J., and Hof, P.R. (2003). Variability of Broca's area homologue in African great apes: implications for language evolution. *Anat. Rec. A Discov. Mol. Cell. Evol. Biol.* 271, 276–285.
- Petrides, M., Tomaiuolo, F., Yeterian, E.H., and Pandya, D.N. (2012). The prefrontal cortex: comparative architectonic organization in the human and the macaque monkey brains. *Cortex* 48, 46–57.
- Petrides, M., and Pandya, D.N. (1994). Comparative cytoarchitectonic analysis of the human and the macaque frontal cortex. In *Handbook of neuropsychology, Volume 9* (Amsterdam: Elsevier), pp. 17–58.
- Paredes, M.F., James, D., Gil-Perotin, S., Kim, H., Cotter, J.A., Ng, C., Sandoval, K., Rowitch, D.H., Xu, D., McQuillen, P.S., et al. (2016). Extensive migration of young neurons into the infant human frontal lobe. *Science* 354, aaf0773.
- Preuss, T.M. (2007). Evolutionary specializations of primate brain systems. In *Primate Origins: Adaptations and Evolution* (Springer), pp. 625–675.
- Striedter, G.F. (2005). *Principles of Brain Evolution* (Sinauer Associates).
- Balsters, J.H., Cussans, E., Diedrichsen, J., Phillips, K.A., Preuss, T.M., Rilling, J.K., and Ramnani, N. (2010). Evolution of the cerebellar cortex: the selective expansion of prefrontal-projecting cerebellar lobules. *Neuroimage* 49, 2045–2052.
- Smaers, J.B., and Soligo, C. (2013). Brain reorganization, not relative brain size, primarily characterizes anthropoid brain evolution. *Proc. Biol. Sci.* 280, 20130269.
- Kozioł, L.F., Budding, D.E., and Chidekel, D. (2012). From movement to thought: executive function, embodied cognition, and the cerebellum. *Cerebellum* 11, 505–525.
- Genovesio, A., Wise, S.P., and Passingham, R.E. (2014). Prefrontal-parietal function: from foraging to foresight. *Trends Cogn. Sci.* 18, 72–81.
- Rilling, J.K., Glasser, M.F., Preuss, T.M., Ma, X., Zhao, T., Hu, X., and Behrens, T.E. (2008). The evolution of the arcuate fasciculus revealed with comparative DTI. *Nat. Neurosci.* 11, 426–428.
- Khabbazian, M., Kriebel, R., Rohe, K., and Ané, C. (2016). Fast and accurate detection of evolutionary shifts in Ornstein–Uhlenbeck models. *Methods Ecol. Evol.* 7, 811–824.
- Ho, L.S.T., and Ané, C. (2014). Intrinsic inference difficulties for trait evolution with Ornstein–Uhlenbeck models. *Methods Ecol. Evol.* 5, 1133–1146.
- Cressler, C.E., Butler, M.A., and King, A.A. (2015). Detecting adaptive evolution in phylogenetic comparative analysis using the Ornstein–Uhlenbeck model. *Syst. Biol.* 64, 953–968.
- Foville, M. (1864). *L'anatomie de la physique et de la pathologie du système nerveux cérébro-spinal* (Paris: Fortin, Masson et Compagnie).

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Supplemental Information

**Exceptional Evolutionary Expansion
of Prefrontal Cortex in Great Apes and Humans**

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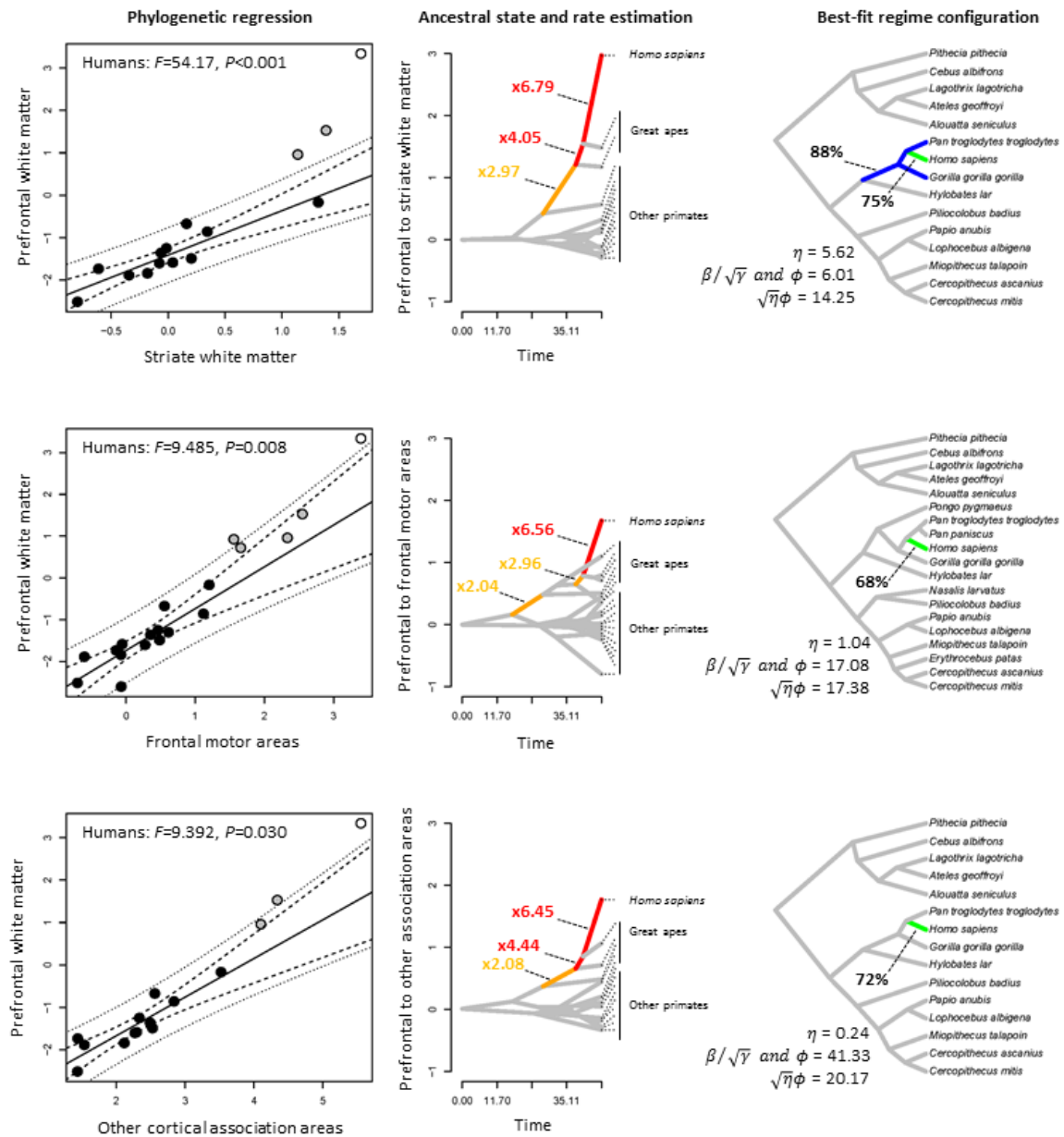


Figure S1: Evolutionary modelling of white matter. Related to Figures 2-3.

As in Figure 2, but for frontal white matter. Data from Smaers and colleagues [S1, S2].

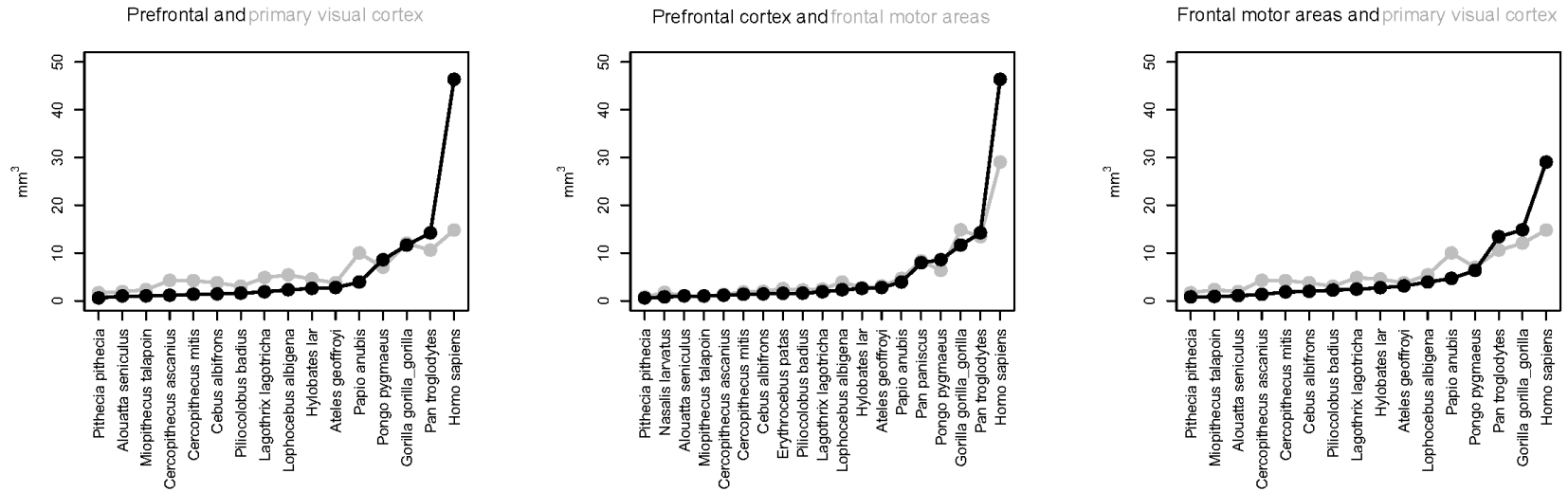


Figure S2: Expansion of prefrontal size in the human lineage. Related to Figures 2 and 4.

As in Figure 4, but for the Smaers data [S1, S2]. We note that, per design, the proxy for prefrontal cortex and frontal motor areas in this data are such that prefrontal cortex volume is underestimated, and frontal motor areas overestimated for great apes and humans. See first section on ‘Data’ in the supplementary information. These proxies thus underestimate trends in prefrontal cortex expansion.

Table S1: **Overview of data.** Related to Figure 1.

Volumes are in mm³, surface area in mm².

Smaers data	Gray matter volume				White matter volume			
	Primary visual	Prefrontal	Other cortical association areas	Frontal motor	Primary visual	Prefrontal	Other cortical association areas	Frontal motor
<i>Cercopithecus ascanius</i>	4.272	1.154	16.060	1.351	0.834	0.159	8.297	0.930
<i>Cercopithecus mitis</i>	4.232	1.380	19.385	1.852	1.042	0.205	9.911	0.946
<i>Erythrocebus patas</i>		1.560		2.552		0.272		1.848
<i>Miopithecus talapoin</i>	2.339	1.018	12.218	0.906	0.709	0.152	4.642	0.548
<i>Alouatta seniculus</i>	1.918	1.015	12.394	1.077	0.543	0.177	4.212	0.866
<i>Ateles geoffroyi</i>	3.753	2.773	32.933	3.119	0.985	0.286	10.332	1.586
<i>Lagothrix lagotricha</i>	4.856	1.887	24.188	2.434	1.226	0.226	12.496	1.625
<i>Cebus albifrons</i>	3.764	1.434	21.024	1.977	0.939	0.257	12.053	1.425
<i>Gorilla gorilla gorilla</i>	12.062	11.642	125.539	14.854	3.123	2.608	60.386	10.295
<i>Homo sapiens</i>	14.787	46.323	271.891	29.013	5.439	28.135	258.218	29.755
<i>Hylobates lar</i>	4.574	2.588	28.424	2.744	1.176	0.510	12.915	1.743
<i>Pan paniscus</i>		7.956		8.356		2.056		5.243
<i>Pan troglodytes troglodytes</i>	10.593	14.209	112.750	13.407	4.004	4.607	76.600	12.780
<i>Pongo pygmaeus</i>	7.008	8.596	76.924	6.353		2.520		4.745
<i>Pithecia pithecia</i>	1.679	0.600	9.242	0.819	0.451	0.081	4.200	0.496
<i>Lophocebus albigena</i>	5.420	2.311	29.102	3.926	1.411	0.424	17.092	3.063
<i>Papio anubis</i>	10.001	3.921	45.240	4.686	3.740	0.845	33.835	3.325
<i>Nasalis larvatus</i>		0.813		1.749		0.074		0.933
<i>Ptilocolobus badius</i>	3.053	1.591	22.629	2.260	0.926	0.201	9.665	1.317

Brodmann data	Surface area			
	Primary visual	Prefrontal	Other cortical association areas	Frontal motor
<i>Callithrix jacchus</i>	341	148	993	167
<i>Saimiri sciureus</i>	744			
<i>Cebus apella</i>	1786	1260	8814	1822
<i>Macaca mulatta</i>	1866	1733	9892	1817
<i>Papio cynocephalus</i>	2559	2111	13506	2200
<i>Papio hamadryas</i>	2130	1967	13599	2898
<i>Mandrillus sphinx</i>	3537	2168	23422	2194
<i>Hylobates lar</i>	1899	1839	10912	1651
<i>Pan troglodytes troglodytes</i>	3212	6719	24252	5389
<i>Homo sapiens</i>	3495	30234	68355	6117

Table S2: **Phylogenetic analysis of covariance.** Related to Figures 2-3.

Results from a phylogenetic analysis of covariance. Results relate to tests of differences in intercept among groups with the slope held constant. ‘Others’ refers to all non-great ape primates in the sample. The analysis includes the comparison of multiple treatment groups (group a ‘v’ group b) to a control group (‘|’ group c). Data from Smaers et al.[S1, S2] and Brodmann [S3].

		Pgls ancova							
		Smaers data				Brodmann data			
Y	X	Grouping	F	P		F	P		
Maximum likelihood estimation of λ									
Prefrontal	Primary visual	Among groups	15.630	<0.001	***	35.728	0.001	**	
		Humans v others great apes v others	30.622	<0.001	***	65.373	<0.001	***	
		Humans v great apes others	13.969	0.003	**	17.087	0.009	**	
		Great apes v others humans	14.431	0.003	**	10.453	0.023	*	
	Frontal motor	Among groups	4.842	0.024	*	24.911	0.003	**	
		Humans v others great apes v others	9.098	0.009	**	49.820	<0.001	***	
		Humans v great apes others	5.050	0.040	*	23.080	0.005	**	
		Great apes v others humans	5.994	0.027	*	2.316	0.189		
	Other association areas	Among groups	3.306	0.072		7.049	0.035	*	
		Humans v others great apes v others	6.389	0.027	*	13.431	0.015	*	
		Humans v great apes others	5.697	0.034	*	3.661	0.114		
		Great apes v others humans	1.931	0.190		6.162	0.056		
	$\lambda=0$								
	Prefrontal	Primary visual	Among groups	19.960	<0.001	***	35.728	0.001	**
			Humans v others great apes v others	32.142	<0.001	***	65.373	<0.001	***
			Humans v great apes others	10.653	0.007	**	17.087	0.009	**
Great apes v others humans			22.342	<0.001	***	10.453	0.023	*	
Frontal motor		Among groups	4.842	0.024	*	24.911	0.003	**	
		Humans v others great apes v others	9.098	0.009	**	49.820	<0.001	***	
		Humans v great apes others	5.051	0.040	*	23.080	0.005	**	
		Great apes v others humans	5.994	0.027	*	2.316	0.189		
Other association areas		Among groups	3.306	0.072		10.395	0.017	*	
		Humans v others great apes v others	6.389	0.027	*	16.525	0.010	*	
		Humans v great apes others	5.697	0.034	*	2.785	0.156		
		Great apes v others humans	1.931	0.190		8.273	0.035	*	
$\lambda=1$									
Prefrontal		Primary visual	Among groups	16.143	<0.001	***	28.728	0.002	**
			Humans v others great apes v others	28.776	<0.001	***	40.753	0.001	**
			Humans v great apes others	23.329	<0.001	***	39.433	0.002	**
	Great apes v others humans		9.180	0.011	**	5.305	0.070		
	Frontal motor	Among groups	4.569	0.028	*	17.291	0.006	**	
		Humans v others great apes v others	7.297	0.016	*	18.145	0.008	**	
		Humans v great apes others	8.438	0.011	*	28.936	0.003	**	
		Great apes v others humans	1.908	0.187		0.989	0.366		
	Other association areas	Among groups	4.417	0.037	*	5.484	0.055		
		Humans v others great apes v others	5.545	0.036	*	10.574	0.023	*	
		Humans v great apes others	8.827	0.012	*	6.210	0.055		
		Great apes v others humans	1.122	0.310		4.236	0.095		

Table S3: **Phylogenetic analysis of covariance.** Related to Figures 1-3.

As in Table S2 considering a maximum likelihood estimation of λ , but for the cumulative prefrontal volume of the 4th and 6th interval along the frontal pole.

<i>Smaers (2011) data, 4th interval</i>			Pgls ancova					
Y	X	Grouping	Gray matter			White matter		
			<i>F</i>		<i>P</i>	<i>F</i>		<i>P</i>
Prefrontal	Primary visual	Among groups	16.354	<0.001	***	25.640	<0.001	***
		Humans v others great apes v others	32.311	<0.001	***	49.399	<0.001	***
		Humans v great apes others	15.526	0.002	**	20.038	0.001	**
		Great apes v others humans	14.292	0.003	**	18.456	0.002	**
	Frontal motor	Among groups	6.371	0.011	*	4.867	0.025	*
		Humans v others great apes v others	11.270	0.005	**	8.808	0.010	*
		Humans v great apes others	5.637	0.032	*	4.916	0.044	*
		Great apes v others humans	8.926	0.010	*	6.724	0.021	*
	Other association areas	Among groups	5.607	0.021	*	5.023	0.031	*
		Humans v others great apes v others	10.977	0.007	**	9.714	0.011	*
		Humans v great apes others	9.347	0.011	*	4.870	0.052	
		Great apes v others humans	3.678	0.081		5.968	0.035	*

<i>Smaers (2011) data, 6th interval</i>			Pgls ancova					
Y	X	Grouping	Gray matter			White matter		
			<i>F</i>		<i>P</i>	<i>F</i>		<i>P</i>
Prefrontal	Primary visual	Among groups	2.925	0.096		28.177	<0.001	***
		Humans v others great apes v others	5.836	0.034	*	54.300	<0.001	***
		Humans v great apes others	3.902	0.074		22.051	<0.001	***
		Great apes v others humans	3.036	0.109		20.252	<0.001	***
	Frontal motor	Among groups	5.054	0.022	*	4.113	0.039	*
		Humans v others great apes v others	9.033	0.009	**	7.549	0.016	**
		Humans v great apes others	4.383	0.055		4.240	0.059	
		Great apes v others humans	6.997	0.019	*	5.862	0.030	*
	Other association areas	Among groups	2.925	0.096		5.074	0.030	*
		Humans v others great apes v others	5.836	0.034	*	9.724	0.011	*
		Humans v great apes others	3.902	0.074		4.682	0.056	
		Great apes v others humans	3.036	0.109		6.262	0.031	*

Table S4: **Testing for the effect of overfitting.** Related to Figures 2-3.

Results from a multi-regime OU model fitting analysis (using the approach proposed by Khabbazzian et al. [S42]) comparing the estimated model (presented in Figures 2 and 3) to an overfit model in which, in addition to the estimated regime configuration, New World monkeys were allocated to a separate regime. Results demonstrate that the overfit model consistently yields a lower pBIC score, confirming that the OU analyses adequately guard against overfitting.

Data	Y	X	pBIC estimated model	pBIC overfit model	Δ pBIC
Smaers data	Prefrontal	Primary visual	26.58	36.47	9.88
		Frontal motor	30.16	37.86	7.71
		Other cortical association areas	-3.49	5.07	8.56
Brodmann data	Prefrontal	Primary visual	19.5	26.12	6.62
		Frontal motor	12.36	18.30	5.93
		Other cortical association areas	15.4	21.49	6.10

Table S5: **Phylogenetic analysis of covariance for white matter.** Related to Figures 2-3.

As in Table S2 considering a maximum likelihood estimation of λ , but for white matter volume instead of gray matter volume. Data from Smaers et al.[S1, S2]

Pgls ancova				
X	Grouping	White matter		
		F	P	
Primary visual	Among groups	28.177	<0.001	***
	Humans v others great apes v others	54.299	<0.001	***
	Humans v great apes others	22.051	<0.001	***
	Great apes v others humans	20.252	<0.001	***
Frontal motor	Among groups	5.055	0.021	*
	Humans v others great apes v others	9.485	0.008	**
	Humans v great apes others	5.570	0.032	*
	Great apes v others humans	6.580	0.022	*
Other cortical association areas	Among groups	4.885	0.030	*
	Humans v others great apes v others	9.392	0.011	*
	Humans v great apes others	4.586	0.055	
	Great apes v others humans	5.952	0.033	*

Table S6: **PGLS regressions parameters**. Related to Figures 2-3.

Slopes and 95% confidence intervals of slopes derived from a phylogenetic generalized least-squares analysis of different subgroups in the sample.

Y	X	PglS						
		Sample	Gray matter			White matter		
			Slope	95% CI		Slope	95% CI	
Prefrontal	Primary visual	All	1.46	0.97	: 1.95	1.90	1.42	: 2.39
		Non-human	1.09	0.67	: 1.51	1.25	0.85	: 1.64
		Non-great apes	0.93	0.54	: 1.31	1.05	0.69	: 1.41
	Frontal motor	All	1.13	1.00	: 1.27	1.35	1.17	: 1.54
		Non-human	1.07	0.92	: 1.22	1.26	1.04	: 1.48
		Non-great apes	0.89	0.63	: 1.15	1.00	0.62	: 1.38
	Other association areas	All	1.18	1.11	: 1.26	1.28	1.12	: 1.43
		Non-human	1.14	1.06	: 1.23	1.18	0.98	: 1.38
		Non-great apes	1.07	0.92	: 1.22	0.94	0.63	: 1.25
	Rest of cortex	All	1.28	1.16	: 1.41	1.37	1.17	: 1.56
		Non-human	1.23	1.08	: 1.37	1.06	0.77	: 1.36
		Non-great apes	1.09	0.80	: 1.38	0.92	0.50	: 1.35
	Rest of brain	All	1.18	1.08	: 1.29	1.58	1.39	: 1.77
		Non-human	1.15	1.02	: 1.28	1.48	1.26	: 1.70
		Non-great apes	1.09	0.81	: 1.37	1.28	0.79	: 1.77

Supplemental Experimental Procedures

Data

Prefrontal and frontal motor data was taken from Smaers et al. [S1, S2] and Brodmann [S3] (Table S1). These data sets differ in the breadth of the comparative sample (19 versus 13 species, respectively), the nature of the measurement (mm^3 versus mm^2), and the cytoarchitectonic criteria that were employed (volumetric bootstrapping along the frontal pole [S2] relative to the cytoarchitectonic border between areas 3 and 4 [S1] versus granular and agranular cortex). Both data sets comprise one individual per species, except for the human data in Smaers [S1, S2] which comprises 8 individuals. Sex is unknown for the Brodmann data set. In the Smaers data, the nonhuman data comprise 7 females and 11 males and the human data comprise 3 females and 5 males. Although some work has suggested sex-by-hemisphere differences in the Smaers data set [S4], no significant sex differences with regards to whole prefrontal cortex volume have been observed.

Only 9 out of 13 species from the Brodmann data set can be used for the current purposes because in this data set not all measurements were taken for all species. The Smaers data set provides only a proxy of prefrontal cortex volume by employing volumetric bootstrapping along the frontal pole. This bootstrapping approach consists of considering the cumulative volume along a set of equidistant intervals along the frontal pole. This approach thus uses the border between area 3 (somatosensory cortex) and area 4 (primary motor cortex) as a functionally and neurobiologically relevant landmark relative to which prefrontal cortex (cumulative volumes posterior to anterior) and frontal motor areas (cumulative volumes posterior to anterior) are defined. This approach hereby accounts for the more caudal position (relative to monkeys) of the border between area 3 and 4 in the great ape and human brain. Any observed enlargement of prefrontal cortex in great apes and humans using this measure is conservative and likely to be an underestimation [S2] because, although validly accounting for the more caudal position of the border between area 3 and 4 in great apes and humans, it does not account for the more caudal position of the border between prefrontal and premotor cortex within the frontal lobe (thus providing an underestimation of prefrontal cortex volume and an overestimation of frontal motor areas in great apes and humans). Previous research using this approach has indicated that scaling patterns of prefrontal cortex volume are equivalent across the frontal pole [S5]. In the current analyses the 5th volumetric interval along the frontal pole was used (Table S2). Table S3 demonstrate that results are equivalent when using the 4th and the 6th volumetric interval.

Data on primary visual cortex was taken from de Sousa et al. [S6]. Other brain data was taken from Stephan et al. [S7]. All data in Smaers et al. [S1, S2], de Sousa et al. [S6] and Stephan et al. [S7] are derived from the same specimens housed at the C&O Vogt Institute for Brain Research [S8]. The phylogenetic tree was taken from Arnold et al. [S9].

Allometry

The extent to which one variable scales with another across a sample of species provides information as to how much one variable is expected to change given one unit of change in another variable. This is quantified by means of the slope of a regression analysis on log transformed data. Two variables are said to scale allometrically when their scaling coefficient is different from unity, and isometrically when it is equivalent to unity. The nature of the allometric relationship among brain structure sizes thus reveals the extent to which changes in one brain structure are expected to outpace the changes in another as overall brain size increases. The scaling coefficient hereby provides information on which changes in proportionality of brain structure sizes are expected given changes in overall brain size. Although traditional scaling studies mostly employed ordinary least squares regression for these purposes, more recent studies use phylogenetic generalized least squares ('pgls') regression to account for potential species dependence due to phylogenetic relatedness [S2, S5]. Specifically, pgls is a generalized least squares regression that uses the expected covariance due to shared phylogenetic history as the covariance matrix among observations in estimating the slope and intercept of a least squares regression [S10-S13].

Functionally and neurobiologically relevant measures of relative prefrontal cortex size

To evaluate whether a particular brain structure is enlarged relative to other structures, the standard approach has been to fit a regression line through a comparative sample and calculate to what extent predicted values correspond to observed values. This approach adequately accounts for scaling differences among different brain structures and is therefore an appropriate measure of relative size (i.e. independently of overall size). This approach was pioneered by

Passingham [S14] using standard least squares regression. Although this approach remains the primary means to assess relative size, more recent studies use phylogenetic generalized least-squares regression [S2, S5, S15].

Relative prefrontal cortex size has commonly been characterized as the size of prefrontal cortex relative to the size of the rest of the cerebral cortex or the rest of the brain [S15]. When aiming to investigate the extent of human prefrontal cortex expansion, this characterization of relative prefrontal cortex size does not, however, account for the functional and anatomical underpinnings of neural information processing [S5]. It has long been established that information processing in the brain occurs in a hierarchical manner. Stimuli are initially mapped onto primary sensory areas, and information subsequently projects to supplementary sensory areas and higher order association areas [S16]. Throughout this process stimuli become increasingly complex [S17] and more object-like [S18]. The increasingly complex information from the stimuli is subsequently integrated in higher order parietal and temporal association areas to form mental representations [S19]. Prefrontal cortex can be considered to stand at the top of this hierarchical mapping of information processing in that it controls the manipulation of, and changes in these mental representations, which, in turn, forms the basis of complex decision making [S20, S21]. The prefrontal cortex is therefore highly connected with other higher order cortical association areas such as parietal and temporal areas [S22, S23]. Therefore, comparing prefrontal cortex to the rest of the cerebral cortex (or rest of the brain) confounds the interpretation of whether prefrontal cortex is enlarged because a comparison is made with a measure that includes both areas with which it is highly integrated and areas with which it is not. A more appropriate comparison embedded within the functional and anatomical framework of neural information processing would be to compare target areas with their input [S5]. In the context of the prefrontal cortex, a comparison with primary sensory areas would reveal whether prefrontal cortex is enlarged relative to its sensory input (effectively comparing higher-order with lower-order information processing), and a comparison with other cortical association areas would reveal whether prefrontal cortex is especially enlarged relative to other heteromodal areas that project to it [S5]. Such results in our analyses unequivocally indicate that both prefrontal cortex and other cortical association areas are significantly expanded in great apes and humans relative to primary sensory areas, and that human prefrontal cortex is significantly expanded relative to other cortical association areas (Table S2, Figures 2-3).

To account for the pathway of information processing we consider four comparisons that together provide a comprehensive analysis of how prefrontal cortex volume has changed relative to other regions: (i) prefrontal cortex size relative to primary visual cortex; (ii) prefrontal cortex size relative to frontal motor areas; (iii) prefrontal cortex size relative to other cortical association areas; (iv) prefrontal cortex size relative to the rest of neocortex.

The comparison relative to the primary visual cortex, which represents the only primary sensory area for which comparative measurements are available, provides the most informative view of changes in prefrontal cortex size because it reliably reflects the hierarchical nature of neural information processing by comparing the amount of 'higher' associative processing capacities relative to the amount of 'lower' primary sensory information processing [S5] (see main text for more information). Similar to previous studies [S5] we consider the size of the primary visual cortex as indicative of the size of other primary sensory areas because it is a measure that is readily available across the primate sample and both have been shown to scale equivalently to body size [S14, S24]. Ideally, all primary sensory areas would be included in this measure, but this information is only available for 5 species of haplorhines [S25]. The fact that not all primary sensory areas are included in these analyses, however, results in a highly conservative test. Because all primary sensory areas are more closely aligned with changes in body size than brain size [S26], changes in visual primary sensory areas align more with other primary sensory areas than they do with cortical association areas.

The comparison of prefrontal cortex with frontal motor areas reflects the hierarchical information processing relationship between primary motor cortex and prefrontal cortex [S21]. However, no volumetric information is available separately on primary motor cortex. Our measure of frontal motor areas therefore includes elements of supplementary motor cortex and premotor cortex [S2, S27]. Because prefrontal and supplementary motor cortex are tightly interconnected [S28], this may confound the evaluation of the putative enlargement of prefrontal cortex volume relative to primary motor cortex. Any such confounding effect will, however, produce conservative results in terms of the expansion of prefrontal cortex relative to primary motor cortex such that any observed trend will be an underestimation.

The comparison of prefrontal cortex with other cortical association areas would reveal whether the prefrontal cortex has enlarged more than other cortical association areas with which it is tightly integrated (primarily higher order

parietal and temporal areas [S22, S23]). The proxy measure for ‘other cortical association areas’ that is used comprises neocortical size minus frontal lobe and primary visual size. Ideally, higher order parietal and temporal areas would be delineated using cytoarchitectonic or tract-tracing techniques, but standards to delineate these areas across species have yet to be developed.

The comparison with the rest of the cerebral cortex is susceptible to the same confounding effect as the comparison with frontal motor areas, but to a much higher degree. The ‘rest of the cortex’ includes both the higher order parietal and temporal areas with which the prefrontal cortex is highly integrated and the lower order primary sensory areas that form the basis of primary sensory information processing. Comparing prefrontal cortex to a measure that includes both higher order association areas and lower order primary sensory areas is functionally uninformative, problematic to interpret neurobiologically, and leads to an unrealistic level of conservatism in estimated patterns of change. More functionally and neurobiologically appropriate comparisons are with separate measures for other cortical association areas and primary sensory areas (as described above).

Phylogenetic analysis of covariance

Before interpreting allometry, formal tests are required to evaluate whether slopes and intercepts are homogenous in all subgroups of the sample [S29]. If such significant differences exist, the allometry may not reflect the most common relationship between size and form in the sample, but rather a departure of a particular subgroup. The consequence of not testing for homogeneity of intercepts and slopes would be that multi-grade isometry is mistaken for one-grade allometry. Indeed, comparing results of pgl regressions through different subsamples demonstrates how overlooking significant intercept differences among subgroups of the sample leads to the illusion that prefrontal cortex scaling is a one-grade allometry instead of a multi-grade isometry (Table S6).

Putative differences in slopes and intercepts among different groups in the sample can be tested using the method of general linear models in which analysis of variance is performed using least-squares regression [S30]. To test for differences in intercepts, we implement phylogenetic analysis of covariance by generalizing the standard generalized least squares procedure ($\mathbf{y} = \mathbf{X}\mathbf{b} + \boldsymbol{\epsilon}$) to include additional indicator variables describing group membership [S31]. In the context of comparing multiple treatment groups to a control group, the GLS regression model here becomes $\mathbf{y} = \mathbf{X}\mathbf{b} + \boldsymbol{\tau}_i + \boldsymbol{\epsilon}$ where $\boldsymbol{\tau}_i$ is the difference from group i to the reference group. $\boldsymbol{\epsilon}$ is hereby assumed to follow a multivariate normal distribution with mean of $\mathbf{0}$ (a null vector of length n) and an $n \times n$ phylogenetic variance-covariance matrix giving the variances and covariances among the observations. Results presented in Table S2 include the λ parameter in order to account for the degree of phylogenetic signal in the data [S12]. Table S2, however, demonstrates that results do not depend on λ (equivalent effects are observed when $\lambda=0$ and $\lambda=1$). Technical details and examples of implementation are available in Smaers & Rohlf [S31].

Phylogenetic analysis of covariance (‘ancova’) calculates the change associated with the clades of interest in the residual variance simultaneously with the phylogenetic regression parameters, thereby accounting for all available variance. This approach allows for a direct test of whether a model with multiple grades (assuming multiple groups with different mean trait values) provides a significantly better fit to the data than a model with only a single grade (assuming that no particular group indicates a significantly different mean trait value). By testing for differences among intercepts, phylogenetic ancova tests for the occurrence of grade shifts in phylogenetic data. It is important to emphasize that the current implementation of phylogenetic ancova uses least-squares procedures only [S31]. Least-squares phylogenetic ancova is hereby no different from standard least-squares ancova with the only exception that the phylogenetic variance-covariance is included in the error term. The advantage of using least-squares procedures is that it produces unbiased results irrespective of sample size. In the least-squares framework low sample size is not a statistical issue because least-squares minimizes the sum of squares of the residuals regardless of sample size and thereby provides unbiased estimates of the regression parameters. Because our implementation of phylogenetic ancova uses least-squares procedures only, this also applies to our analyses. Small sample size may of course, present empirical issues, though this is not the case for our analyses.

Least-squares phylogenetic ancova is preferred over methods that detect ‘evolutionary singularities’ [S32, S33] because such methods do not provide a direct test for differences in intercepts and slopes in a regression, nor do they provide a direct test comparing the fit of multiple grades relative to a single grade. Phylogenetic ancova is also preferred over the previously used procedure to remove the datum of interest, fit a regression line to the remaining

data points, and calculate prediction intervals for a new observation [S15, S34]. Such a procedure is not a correct statistical test because it does not account for the variance of the predicted means [S31].

Furthermore, because the procedure for phylogenetic ancova used here employs standard least-squares procedures only [S31], parameterization is automatically penalized within the estimation of the probabilities. Based on personal communications some have suggested that at least 10 data points are needed per parameter to be estimated [S35]. However, this cut-off is trivial and is suggested to apply to multidimensional data. Considering that our data are bivariate only, and that calculations are done using standard least-squares procedures only (which automatically penalize parameterization in the estimation of the probabilities), the parameterization of two or three grades is acceptable and poses no statistical issue.

Given the small sample size of the Brodmann data it may be purported that least-squares procedures do not adequately penalize for the added parameters assumed by including multiple groups in analyses. This question relates to the notion of whether degrees of freedom penalize for parameters in the least-squares framework in such a way to avoid overfitting. Considering that least-squares analysis is the golden standard in regression analysis, there is overwhelming support to accept this notion. We put this to the test for our data by testing whether a 3-grade model (human v great ape v other primates) does indeed provide a better fit than a 2-grade model (human v nonhuman), and whether a 4-grade model (human v great ape v other Old World primates v New World primates) provides a better fit than a 3-grade model. Note that an F-ratio test only tests whether a more parameterized model provides a significantly *better* fit than a less parameterized model (not vice versa). If the procedure tends to overfit, then a more parameterized model would always provide a significantly higher fit than a less parameterized model because the addition of more parameters would not be adequately penalized.

We tested this on the prefrontal-primary visual data. For both the Brodmann and Smaers data, we found that a 3-grade model does indeed provide a better fit than a 2-grade model (Brodmann: $F=10.4532$, $P=0.0231$; Smaers: $F=14.4314$, $P=0.0025$). This confirms our conclusion that both great apes and humans can be considered separate non-allometrically derived grades. Importantly, a 4-grade model is not significantly different from a 3-grade model (Brodmann: $F=0.9569$, $P=0.3834$; Smaers: $F\sim 0$, $P=0.9785$), demonstrating that least-squares phylogenetic ancova avoids overfitting in our analyses.

An assumption of the use of phylogenetic ancova to test for differences in intercepts among subgroups is that slopes are homogenous. The difference in the slope of the regression lines between the non-great ape sample and the great ape sample (including humans) is not larger than expected by chance for the Smaers data (Prefrontal vs primary visual: $F=2.712$, $P=0.126$; Prefrontal vs frontal motor: $F=0.911$, $P=0.355$; Prefrontal vs other cortical association areas: $F=2.263$, $P=0.158$). The Brodmann data set does indicate such differences (Prefrontal vs primary visual: $F=17.087$, $P=0.009$; Prefrontal vs frontal motor: $F=23.080$, $P=0.005$; Prefrontal vs other cortical association areas: $F=3.661$, $P=0.114$). The great ape sample in the Brodmann data set, however, consists of only 2 species (chimpanzee and human). Considering that scaling relationships among different variables and subgroups are equivalent in the Smaers and Brodmann data sets [S5] it is reasonable to assume that the difference in slopes in the Brodmann data set are due to $n=2$ in the great ape and human sample rather than representative of the true difference in slope (for which the Smaers data set gives a better estimate considering its larger comparative sample). Moreover, when a difference in slope is observed among subgroups, it can already be concluded that the subgroups belong to a different allometry (in such a case there is no need for a test of differences in intercepts to conclude that subgroups are part of a different grade). The interpretation of a phylogenetic ancova test for the Brodmann data set is therefore valid and provides a conservative estimate.

Evolutionary modelling

To provide additional tests for the occurrence of grade shifts we performed a series of different evolutionary modelling analyses. Using a phylogenetic tree and observed information from contemporary tip taxa, these evolutionary modelling approaches employ statistical and mathematical models of evolution to describe the pattern and rate of trait change along individual branches of a phylogeny. Specifically, we use Ornstein-Uhlenbeck ('OU') model fitting, and ancestral state and rate estimation methods. Each of these methods assume different models of evolution. OU model fitting assumes a multi-regime OU model. Ancestral state and rate estimation methods were used assuming two different multi-rate models (a multiple variance Brownian motion and a reversible-jump BM model), both of which produced equivalent results. These different methods add to our analyses by elucidating when in evolutionary time

deviations from allometric conservation occurred, and by demonstrating that the same results are obtained using different methods and different assumed models of evolution.

In all evolutionary modelling analyses, we calculated the relative size of prefrontal cortex for great apes and humans as the difference between the expected values based on the non-great ape sample relative to their observed values. We model the evolutionary expansion of prefrontal cortex volume in this manner for three reasons: (1) In order to investigate whether human prefrontal cortex follows general primate brain allometry, previous work [S15, S34] excluded the human value from the analyses and compared the predicted human value with the observed human value by assessing whether the latter lies within the confidence intervals of the former. In other words, the human value was assessed relative to the non-human allometry. Although the idea to exclude the human value of interest to assess whether the non-human data aligns with non-human allometry is valid, the approach of evaluating whether the observed value lies within or outside a confidence/prediction interval constitutes an incorrect test because it does not take into account the expected variance of the predicted means [S31]. Our approach to consider great apes and humans relative to the non-great ape/human pattern hereby mirrors the approach of previous work, but rather than performing an incorrect test on these values, we model these values in a manner that accounts for all variance. (2) Phylogenetic analysis of covariance unequivocally demonstrates that humans and great apes do not follow the same allometry as other primates (Table S2). This demonstrates that a multi-grade model is strongly preferred over a one grade model and therefore that residuals should be taken using a multi-grade model, not a one grade model. (3) The nature of the phylogenetic relationship between non-great ape primates and great apes (great apes are embedded within the earlier radiation of monkeys, see Figure 2) indicates that whatever trend is prevalent across monkeys is the ancestral condition of great apes. In order to assess the evolutionary expansion of great apes and humans the most appropriate test is thus to model their expansion relative to the ancestral non-great ape sample [S5, S36].

Evolutionary modelling: OU model fitting

The most frequently used statistical model of evolution is standard Brownian motion ('BM'), which assumes that traits change at each unit of time with a mean change of zero and unknown and constant variance [S37-S39]. Within Brownian motion, the evolution of a continuous trait 'X' along a branch over time increment 't' is quantified as $dX(t) = \sigma dB(t)$, where ' σ ' constitutes the magnitude of undirected, stochastic evolution (' σ^2 ' is generally presented as the Brownian rate parameter) and ' $dB(t)$ ' is Gaussian white noise. It is paramount to consider that, at a fundamental level, BM quantifies the accumulation of *variance* over time. Although novel phylogenetic comparative methods continue using BM as a baseline model, they incorporate additional parameters to model changes in *mean* value over time. By incorporating additional parameters to quantify differences in mean value, these more complex models (Ornstein-Uhlenbeck models, or 'OU' models) quantify possible deviations from the standard gradual mode of evolution assumed by BM. Specifically, OU models extend the standard BM model to include a parameter that represents the mean trait value (θ) and the rate at which changes in mean values are observed (α). The OU model hereby assumes that the evolution of a continuous trait 'X' along a branch over time increment 't' is quantified as $dX(t) = \alpha[\theta - X(t)]dt + \sigma dB(t)$ [S40]. The inclusion of parameters that estimate mean trait value (θ) and the rate at which changes in mean values are observed (α) renders the OU model framework more appropriate than BM for modelling changes in the direction of trait evolution. This is true because the OU model adequately differentiates between changes in the mean (θ and α) and variance (σ^2) of a trait over time. This standard OU model has been modified into multiple-regime OU models that allow modelling trait evolution towards different 'regimes' that each display a different mean trait value [S41]. Several methods have been developed that use this approach to model trait diversification by estimating shifts in θ values along the branches of the phylogeny [e.g., S42, S43].

Although OU modelling approaches are commonly agreed to be very powerful for modelling trait diversification, recent research has pointed towards some challenges in traditional methods. Specifically, the theoretical properties of the maximum-likelihood estimators for OU parameters can result in non-uniqueness and inaccuracy, and traditional model selection criteria tend to favor overly complex scenarios [S44]. More recent approaches, however, overcome these difficulties by using a least-squares procedure and modifying model selection criteria so that they guard against overfitting [S42, S44]. Khabbazian et al. [S42], for example, developed a least-squares procedure to detect shifts on individual branches and a phylogenetic lasso procedure to regularize the shift configuration across the tree. This procedure ensures statistical selection consistency, small prediction error and uniqueness of estimators. To avoid overfitting, this procedure further employs a phylogenetic Bayesian Information Criterion ('pBIC') that better estimates model posterior probability than the traditional Akaike Information Criterion ('AIC') when errors are phylogenetically correlated. This procedure has been demonstrated to be highly conservative in that it provides high

precision (the detected shifts are true) but a low recall rate (some true shifts are missed) [S42]. We use Khabbazian et al.'s approach in our analysis.

Given the small sample size of the Brodmann data it may also be purported that the analysis would not adequately penalize for the added parameters assumed by including multiple groups. This assertion is counter to evidence that pBIC reliably guards against overfitting [S42], but may deserve special scrutiny for our particular data set. To test putative overfitting, we compared the 'estimated model' (displayed in Figures 2 and 3) against a more parameter-rich model in which, in addition to the estimated regime configuration, New World monkeys are allocated to a separate regime ('overfit model'). If our analyses were prone to overfitting, the more highly parameterized overfit model would produce a better statistical fit to the data than the estimated model. We find, however, that, for each analysis for both the Smaers and Brodmann data, the estimated model provides a better fit than the overfit model ($\Delta\text{pBIC} > 5$: see Table S4). These results demonstrate that also our OU analyses are robust against overfitting.

Evolutionary modelling: Ancestral state and rate estimation

To visualize trait change over time we use multi-rate Brownian motion models to estimate values at internal nodes of the phylogenetic tree and rates of evolution of particular traits across the entire sample. Traditionally, standard BM models have been used for these purposes [S45]. A consequence of assuming standard BM is that average trait change is proportional to the square root of time and that the rate of change is constant such that it has a single mean and variance across all branches. There is, however, evidence to suggest that the direction and rate of some traits may change among different lineages [S46]. In such instances, the standard BM rate parameter is not expected to capture all meaningful evolutionary trait variation. To account for lineage-specific variation in direction and rate of evolution, recent methods have expanded the standard BM model to include the estimation of multiple rates along the phylogeny. Here we compare results of the reversible-jump MCMC procedure ('rjBM') introduced by Venditti et al [S47, S48], and the multiple variance BM ('mvBM') procedure introduced by Smaers et al. [S49]. Because estimated rates in the rjBM procedure can be interpreted as traditional standard BM rate parameters (σ^2), the multi-rate rjBM model can be parameterized using a standard BM model to estimate ancestral states. More information on this procedure and its implementation is available from the manual of the software package that supports this procedure [S48]. The MCMC procedure uses 10 million iterations, sampling every 100th iteration, ensuring normal distribution of log likelihood values.

Using the rjBM and mvBM procedures we further calculated the extent to which the rjBM and mvBM rate estimates for individual lineages differ from a baseline expectation of gradual evolution. The amount of change observed at each branch (calculated as the difference between ancestral values as inferred from mvBM or rjBM approaches and descendent values) was hereby compared with a neutral scenario in which all the species in the phylogeny were simulated to evolve at a constant rate [S50]. The original phylogeny was rescaled to generations by dividing each branch length (in Myrs) by the generation time corresponding to their descendent species or node. For internal nodes, ancestral generation times were calculated using a constant variance BM approach. Age at first reproduction obtained from van Schaik & Isler [S51], and missing values from the PanTHERIA database [S52], was used as a proxy for generation time because this parameter is available for all the species in our dataset. Once the phylogeny was rescaled in this way, a per-generation evolutionary rate (σ^2) was calculated based on each dataset. This constant rate was used to simulate evolutionary change across the phylogeny as a random walk with branching points at corresponding nodes. Simulations were run 1,000 times and the mean change between all ancestors and descendants was used as the expectation of the amount of change had each branch evolved neutrally under a pure BM model. rjBM- and mvBM-based measures of change per branch were compared with mean simulated amounts of change per branch by calculating a ratio between the former and the latter. A ratio greater than 1 indicates that a species has experimented a greater amount of evolutionary change than expected in a constant-rate neutral scenario regardless of the directionality of change, whereas a ratio smaller than 1 indicates that a species has changed less than expected. For the present data set, rjBM and mvBM approaches yielded similar results. Analyses were carried out in Mathematica 10 using the package 'Phylogenetics' [S53].

Power, effect size, and sample size.

Given the small sample size in the Brodmann data, it is important to clarify how such low sample size relates to the power, effect size, and thus reliability of our results.

The power of a test determines the probability of rejecting the null hypothesis when it is in fact false. In general, the issue of power only arises when results are *not* significant. The power of a test therefore helps determine the sample size that is needed in future experiments to reliably discriminate between the null hypothesis and the alternative hypothesis. The calculation of power thus relates to the research design of future experiments. One can also calculate the power after an experiment has already been concluded. In this case, the power is referred to as the *observed power*. This case relates to the analyses we present. But here also, the issue of observed power only arises when results are not significant. When the observed power is high in a non-significant test, this is sometimes used to argue in support of the null hypothesis. This interpretation is, however, inappropriate and is referred to as the “power approach paradox” [S54]. When results are, however, significant (as is the case in our results), the computation of observed power does not add information over the *P*-value because the observed power is a simple function of the observed *P*-value [S54]. In other words, when the *P*-value indicates that the test is significant, the observed power of the test is, per definition, high. Considering that our ancova results are significant, the observed power of our tests must be high.

Furthermore, as mentioned, the phylogenetic ancova procedure we employ uses least-squares procedures only. Least-square procedures minimize the sums of squares of the residuals and hereby always provide unbiased estimates of the regression parameters regardless of sample size. The fact that low sample size is not a statistical issue for least-squares analysis should not, however, be confounded with the uncertainty that low sample size causes for the maximum likelihood estimation of parameters of models of evolution. The estimation of the λ parameter, for example, has been shown to have adequate power only for trees larger than 20 tips [S55]. The inclusion of the λ parameter in phylogenetic least-squares analysis thus introduces an element of uncertainty. We show that for our analysis, results are not dependent on the λ parameter (equivalent results are obtained in both extreme cases $\lambda=0$ and $\lambda=1$, see Table S2). Because our results are not dependent on the λ parameter, they are also not dependent on the uncertainty due to low sample size in estimating the λ parameter. It is paramount that the uncertainty in the maximum likelihood estimation of λ due to small sample size is not confounded with the purported uncertainty in the least-squares calculation of the regression parameters. As mentioned, least-squares procedures provide unbiased calculations of regression parameters irrespective of sample size.

With respect to uncertainty due to low sample size in the estimation of parameters of models of evolution it is further crucial to adequately differentiate between different models of evolution. Different models make different assumptions about the evolutionary process, and thereby include different statistical parameters that represent those assumptions. As mentioned, the standard BM model of evolution assumes the evolution of a continuous trait ‘*X*’ along a branch over time increment ‘*t*’ is quantified as $dX(t) = \sigma dB(t)$. Considering that σ represents the dispersion of the normal distribution from which changes per unit of time are sampled, it is evident that a low sample size will hamper its reliable estimation. In other words, low sample size is expected to limit the reliable estimation of the one parameter that is assumed by a standard BM model in this regard (i.e. σ).

The case of the OU model (used in our analysis of best-fit regime configurations) should, however, be differentiated from the case of the standard BM model. As mentioned, the standard BM model quantifies *variance* (σ^2) over time whereas the OU model includes additional parameters (θ and α) to quantify changes in *mean* values over time. Crucially, the inclusion of these two extra parameters also influences the evaluation of power. Within an OU model, power is not dependent only on σ (as in BM), but also on θ and α . In this regard, several authors have demonstrated that power within an OU modelling framework is, in fact, better predicted by effect size than by sample size [S44, S56]. Several measures of effect size have been proposed. Ho & Ané [S57] proposed $\beta/\sqrt{\gamma} > 2$ as indicating high effect size. β hereby indicates the difference in mean trait value among regimes (or $\Delta\theta$), and γ represents the stationary variance ($\sigma^2/(2\alpha)$). Ho & Ané [S57] found that their measure of effect size is a better predictor of power than sample size. Similarly, Cressler et al. [S56] found that combining a measure of selection opportunity $\eta = \alpha T$ (where *T* is the total depth of the phylogeny) with an equivalent measure of effect size $\phi = \sqrt{2\eta}/\gamma = \sqrt{2\alpha}\Delta\theta/\sigma$ into a ‘signal-to-noise’ ratio ($\sqrt{\eta}\phi$) provided a better prediction of power than sample size (high power can be inferred when $\sqrt{\eta}\phi \gg 1$). Because OU modelling approaches primarily aim to detect shifts in *mean* values over time, OU modelling power thus needs to be addressed primarily in terms of effect size. If effect size is high (difference in mean trait values among groups is high), the power to discriminate among different groups is high. Within the OU model framework, power thus crucially depends on the magnitude of the difference in mean trait values among groups. In all our analyses, results demonstrate that effect size is exceptionally high. In other words, differences between humans, great apes, and other primates are so pronounced that alternative regime configurations have a low statistical fit.

Effect size can be put to the test using bootstrap analysis. Bootstrap analysis includes fewer assumptions than estimating regime configurations directly from the data, and can thus be considered as a conservative approach. Khabbazian et al [S42] propose a bootstrap analysis for OU modelling that consists of randomizing residuals across the tree, estimating shift configurations for each iteration, and calculating the proportion of bootstrap iterations when a shift was detected on a particular branch. Results are presented in Figures 2-3 and Figure S1. Bootstrap results further confirm the exceptionally high effect size observed in our data.

An additional way in which effect size can be estimated is to test the same hypothesis using different methods with different model assumptions. If the same result is obtained across different methods and model assumptions, it can be concluded that the effect size is high regardless of the methods used and models assumed. We followed this principle in our research design by testing the hypothesis of significant differences among humans, great apes and other primates in prefrontal expansion using four different methods and five different model assumptions. The four methods include phylogenetic ancova, ancestral estimation, rate estimation, and OU model fitting. The five model assumptions include standard BM [S10] and λ -adjusted BM [S12] (phylogenetic ancova), reversible-jump BM [S47] and multiple variance BM [S49] (ancestral and rate estimation), and multi-regime OU [S42] (OU model fitting). All approaches converge on the same conclusion that humans, great apes, and other primates are distinct grades in prefrontal expansion.

In general, although there is no statistical reason to doubt the reliability and validity of our results, it is clear that a larger sample size would further increase statistical confidence in the results of the analyses. We hope that the present results inspire future studies to continue to expand comparative neuroanatomical data sets.

Supplemental References

- S1. Smaers, J.B., Schleicher, A., Zilles, K., and Vinicius, L. (2010). Frontal white matter volume is associated with brain enlargement and higher structural connectivity in haplorrhine primates. *PLoS One* 5, e9123.
- S2. Smaers, J.B., Steele, J., Case, C.R., Cowper, A., Amunts, K., and Zilles, K. (2011). Primate prefrontal cortex evolution: human brains are the extreme of a lateralized ape trend. *Brain, Behavior and Evolution* 77, 67-78.
- S3. Brodmann, K. (1909). *Vergleichende lokalisationslehre der grosshirnrinde in ihren prinzipien dargestellt auf grund des zellenbaues*, (Leipzig: Barth).
- S4. Smaers, J.B., Mulvaney, P.I., Soligo, C., Zilles, K., and Amunts, K. (2012). Sexual dimorphism and laterality in the evolution of the primate prefrontal cortex. *Brain, Behavior and Evolution* 79, 205-212.
- S5. Passingham, R.E., and Smaers, J.B. (2014). Is the prefrontal cortex especially enlarged in the human brain? *Brain, behavior and evolution* 84, 156-166.
- S6. de Sousa, A.A., Sherwood, C.C., Schleicher, A., Amunts, K., MacLeod, C.E., Hof, P.R., and Zilles, K. (2010). Comparative cytoarchitectural analyses of striate and extrastriate areas in hominoids. *Cerebral Cortex* 20, 966-981.
- S7. Stephan, H., Frahm, H., and Baron, G. (1981). New and revised data on volumes of brain structures in insectivores and primates. *Folia Primatologica* 35, 1-29.
- S8. Zilles, K., Amunts, K., and Smaers, J.B. (2011). Three brain collections for comparative neuroanatomy and neuroimaging. In *Resources and Technological Advances for Studies of Neurobehavioral Evolution*, Volume 1225, J.I. Johnson, H.P. Zeigler and P.R. Hof, eds. (New York: Annals of the New York Academy of Sciences), pp. E94-E104.
- S9. Arnold, C., Matthews, L.J., and Nunn, C.L. (2010). The 10k trees website: A new online resource for primate phylogeny. *Evolutionary Anthropology* 19, 114-118.
- S10. Grafen, A. (1989). The phylogenetic regression. *Philosophical Transactions of the Royal Society of London Series B-Biological Sciences* 326, 119-157.
- S11. Martins, E.P., and Hansen, T.F. (1997). Phylogenies and the comparative method: A general approach to incorporating phylogenetic information into the analysis of interspecific data. *American Naturalist* 149, 646-667.
- S12. Pagel, M. (1997). Inferring evolutionary processes from phylogenies. *Zoologica Scripta* 26, 331-348.
- S13. Rohlf, F.J. (2001). Comparative methods for the analysis of continuous variables: Geometric interpretations. *Evolution* 55, 2143-2160.
- S14. Passingham, R.E. (1973). Anatomical differences between the neocortex of man and other primates. *Brain, Behavior and Evolution* 7, 337-359.
- S15. Barton, R.A., and Venditti, C. (2013). Human frontal lobes are not relatively large. *Proceedings of the National Academy of Sciences* 110, 9001-9006.
- S16. Kaas, J.H. (1997). Topographic maps are fundamental to sensory processing. *Brain research bulletin* 44, 107-112.
- S17. Desimone, R., and Schein, S.J. (1987). Visual properties of neurons in area V4 of the macaque: sensitivity to stimulus form. *Journal of neurophysiology* 57, 835-868.
- S18. Tanaka, K. (1993). Neuronal mechanisms of object recognition. *Science* 262, 685-685.
- S19. Tsunoda, K., Yamane, Y., Nishizaki, M., and Tanifuji, M. (2001). Complex objects are represented in macaque inferotemporal cortex by the combination of feature columns. *Nature neuroscience* 4, 832-838.
- S20. Fuster, J.M. (2008). *The prefrontal cortex*, 4th Edition, (Boston, MA: Academic Press/Elsevier).

- S21. Miller, E.K., and Cohen, J.D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience* 24, 167-202.
- S22. Paulus, M.P., Hozack, N., Zauscher, B., McDowell, J.E., Frank, L., Brown, G.G., and Braff, D.L. (2001). Prefrontal, parietal, and temporal cortex networks underlie decision-making in the presence of uncertainty. *Neuroimage* 13, 91-100.
- S23. Genovesio, A., Wise, S.P., and Passingham, R.E. (2014). Prefrontal–parietal function: from foraging to foresight. *Trends in cognitive sciences* 18, 72-81.
- S24. Passingham, R.E. (1975). Changes in size and organization of brain in man and his ancestors. *Brain Behavior and Evolution* 11, 73-90.
- S25. Shariff, G.A. (1953). Cell counts in the primate cerebral cortex. *Journal of Comparative Neurology* 98, 381-400.
- S26. Passingham, R.E. (1981). Primate specializations in brain and intelligence. *Symp Zool Soc Lond* 46, 361-388.
- S27. Smaers, J.B., Steele, J., Case, C.R., and Amunts, K. (2013). Laterality and the evolution of the prefronto-cerebellar system in anthropoids. In *Evolution of Human Handedness*, W.C. McGrew, L. Marchant and W. Schiefenhövel, eds. (New York: Annals of the New York Academy of Sciences).
- S28. Carmichael, S., and Price, J.L. (1995). Sensory and premotor connections of the orbital and medial prefrontal cortex of macaque monkeys. *Journal of Comparative Neurology* 363, 642-664.
- S29. Sokal, R., and Rohlf, F. (2012). *Biometry*. 4th Edition. (New York, USA: WH Freeman).
- S30. Rao, C., and Toutenburg, H. (1999). *Linear models: least squares and alternatives*, 2nd Edition, (New York: Springer).
- S31. Smaers, J., and Rohlf, F. (2016). Testing species' deviation from allometric predictions using the phylogenetic regression. *Evolution* 70, 1145-1149.
- S32. Nunn, C.L., and Zhu, L. (2014). Phylogenetic prediction to identify “evolutionary singularities”. In *Modern Phylogenetic Comparative Methods and Their Application in Evolutionary Biology*. (Springer), pp. 481-514.
- S33. Organ, C., Nunn, C.L., Machanda, Z., and Wrangham, R.W. (2011). Phylogenetic rate shifts in feeding time during the evolution of Homo. *Proceedings of the National Academy of Sciences* 108, 14555-14559.
- S34. Gabi, M., Neves, K., Masseron, C., Ribeiro, P.F., Ventura-Antunes, L., Torres, L., Mota, B., Kaas, J.H., and Herculano-Houzel, S. (2016). No relative expansion of the number of prefrontal neurons in primate and human evolution. *Proceedings of the National Academy of Sciences*, 201610178.
- S35. Freckleton, R.P., and Watkinson, A.R. (2001). Nonmanipulative determination of plant community dynamics. *Trends in Ecology & Evolution* 16, 301-307.
- S36. Smaers, J.B. (2013). How humans stand out in frontal lobe scaling. *Proceedings of the National Academy of Sciences* 110, E3682.
- S37. Cavalli-Sforza, L.L., and Edwards, A.W. (1967). Phylogenetic analysis. Models and estimation procedures. *Am. J. Hum. Genet.* 19, 233.
- S38. Felsenstein, J. (1973). Maximum-likelihood estimation of evolutionary trees from continuous characters. *Am. J. Hum. Genet.* 25, 471.
- S39. Felsenstein, J. (1985). Phylogenies and the comparative method. *Am. Nat.* 125, 1-15.
- S40. Hansen, T.F. (1997). Stabilizing selection and the comparative analysis of adaptation. *Evolution* 51, 1341-1351.
- S41. Butler, M.A., and King, A.A. (2004). Phylogenetic comparative analysis: A modeling approach for adaptive evolution. *American Naturalist* 164, 683-695.

- S42. Khabbazian, M., Kriebel, R., Rohe, K., and Ané, C. (2016). Fast and accurate detection of evolutionary shifts in Ornstein–Uhlenbeck models. *Methods in Ecology and Evolution* 7, 811-824.
- S43. Uyeda, J.C., and Harmon, L.J. (2014). A novel bayesian method for inferring and interpreting the dynamics of adaptive landscapes from phylogenetic comparative data. *Systematic Biology* 63, 902-918.
- S44. Ho, L.S.T., and Ané, C. (2014). Intrinsic inference difficulties for trait evolution with Ornstein-Uhlenbeck models. *Methods in Ecology and Evolution* 5, 1133-1146.
- S45. Harvey, P.H., and Pagel, M. (1991). *The comparative method in evolutionary biology*, (New York: Oxford University Press).
- S46. Simpson, G.G. (1944). *Tempo and mode in evolution*, (Columbia University Press).
- S47. Venditti, C., Meade, A., and Pagel, M. (2011). Multiple routes to mammalian diversity. *Nature* 479, 393-396.
- S48. Pagel, M., and Meade, A. (2013). *BayesTraits v. 2.0*. Reading: University of Reading.
- S49. Smaers, J.B., Mongle, C.S., and Kandler, A. (2016). A multiple variance Brownian motion framework for the estimation of ancestral states and rates of evolution. *Biological Journal of the Linnean Society* 118, 78-94.
- S50. Gómez-Robles, A., Smaers, J.B., Holloway, R.L., Polly, P.D., and Wood, B. (In Press). Brain enlargement and dental reduction were not linked in hominin evolution. *Proceedings of the National Academy of Sciences*.
- S51. van Schaik, C.P., and Isler, K. (2012). Life-history evolution in primates. In *The Evolution of Primate Societies*, J.C. Mitani, J. Call, P.M. Kappeler, R. Palombit and J.S. Silk, eds. (University of Chicago Press), pp. 220-244.
- S52. Jones, K.E., Bielby, J., Cardillo, M., Fritz, S.A., O'Dell, J., Orme, C.D.L., Safi, K., Sechrest, W., Boakes, E.H., and Carbone, C. (2009). PanTHERIA: a species-level database of life history, ecology, and geography of extant and recently extinct mammals. *Ecology* 90, 2648-2648.
- S53. Polly, P.D. (2012). *Phylogenetics for mathematica*. (Bloomington, IN: Department of Geological Sciences, Indiana University. See <http://mypage.iu.edu/~pdpolly/Software.html>).
- S54. Hoenig, J.M., and Heisey, D.M. (2001). The abuse of power: The pervasive fallacy of power calculations for data analysis. *The American Statistician* 55, 1-6.
- S55. Freckleton, R.P., Harvey, P.H., and Pagel, M. (2002). Phylogenetic analysis and comparative data: A test and review of evidence. *American Naturalist* 160, 712-726.
- S56. Cressler, C.E., Butler, M.A., and King, A.A. (2015). Detecting adaptive evolution in phylogenetic comparative analysis using the Ornstein-Uhlenbeck model. *Systematic biology*, syv043.
- S57. Ané, C. (2014). A linear-time algorithm for Gaussian and non-Gaussian trait evolution models. *Systematic Biology* 63, 397-408.